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48

THE COPPERBELT UNIVERSITY

MCS SCHOOL OF MEDICINE

Test-1 - Human Physiology, Topics: Blood, CVS

TOTAL MARKS = 100

DURATION = 2 hours

SIN Number.....

Section-A, MCQ Answer All in this same page.

Marks = 8.

1. Which of the following best describes the relationship between hematocrit, hemoglobin, and red blood cells in the context of red blood cell indices?
 - A) Hematocrit and hemoglobin are unrelated to red blood cell indices.
 - B) Red blood cell indices are directly measured without involving hematocrit or hemoglobin.
 - ☒ C) Red blood cell indices like MCV, MCH, and MCHC are calculated using mathematical formulas involving hematocrit, hemoglobin, and red blood cell counts.
 - D) Hemoglobin concentration alone determines all red blood cell indices.
2. Which of the following can cause macrocytic anemia?
 - A) Iron deficiency
 - ☒ B) Vitamin B-12 deficiency
 - C) Chronic kidney disease
 - D) Dehydration
3. Which formula correctly represents the calculation of Mean Corpuscular Hemoglobin (MCH)?
 - A) $MCH = \text{Hematocrit (\%)} \div \text{RBC count}$
 - ☒ B) $MCH = \text{Hemoglobin (g/dL)} \times 10 \div \text{RBC count}$
 - C) $MCH = \text{Hemoglobin (g/dL)} \div \text{Hematocrit (\%)}$
 - D) $MCH = \text{RBC count} \div \text{Hemoglobin (g/dL)} \times 10$
4. Which of the following oral manifestations may be associated with anemia?
 - ☒ A) Glossodynia
 - B) Tooth decay
 - C) Gingival overgrowth
 - D) Oral lichen planus

SECTION - B- MTF Answer T/F for each option below Marks = 40

1. Which of the following statements about T lymphocyte maturation in the thymus are true or false?
- a) T lymphocytes mature in the cortex of the thymus. **T**
 - b) T lymphocytes are exposed to antigens in the medulla of the thymus. **T**
 - c) Mature T lymphocytes enter the bloodstream directly from the cortex. **F**
 - d) The thymus gland is responsible for the maturation and selection of T lymphocytes. **T**

2. Which of the following statements about pluripotent and totipotent stem cells are true or false?
- a) Pluripotent stem cells can develop into any cell type in the body but cannot form a complete organism. **F**
 - b) Totipotent stem cells can develop into any cell type and can form a complete organism. **T**
 - c) Pluripotent stem cells are capable of forming extraembryonic tissues like the placenta. **F**
 - d) Totipotent stem cells have a more limited differentiation potential compared to pluripotent stem cells. **F**

3. Which of the following statements about erythropoietin production are true or false?

- a) Renal erythropoietic factor and liver erythropoietinogen combine to form erythropoietin in the blood. **T**
- b) The kidney is the primary source of erythropoietin production in adults. **T**
- c) The liver is the major site of erythropoietin production during fetal development. **T**
- d) Erythropoietin is produced exclusively by the liver in adult humans. **F**

4. For each of the following components of an ECG, indicate whether the matching event occurring in the heart is true or false:

- a) P wave - atrial depolarization **T**
- b) PR interval - AV node absent **F**
- c) QRS complex - ventricular depolarization **T**
- d) T wave - atrial repolarization **T**

5. For each of the following statements regarding the significance of the Frank-Starling law of the heart, indicate whether it is true or false:

- a) The more the heart fills with blood during diastole, the greater the contraction force. **T**
- b) The less the heart fills with blood, the greater the contraction force. **F**
- c) Greater sympathetic stimulation results in higher contraction force. **T**
- d) Greater sympathetic stimulation results in a shorter relaxation period. **T**

Section - C - Structured Questions
Each = 4

1. Explain exam

Section – C – Structured Questions - Answer any SEVEN in the white sheets, Marks = 28**Each = 4**

1. Explain operational effect of Starling forces on capillary physiology, with examples of numbers/values/diseases.
2. Using an ECG diagram, explain the possible causes and ECG changes associated with substernal chest pain in a patient.
3. What precautions and considerations are necessary when prescribing antiplatelet, anticoagulant, and antihypertensive medications to dental patients with a history of cardiovascular disease? Give reasons.
4. When ever a person stands up from supine position, the blood pressure drops initially and then comes back to normal. Explain this physiology concept.
5. Explain stages of erythropoiesis with labelled diagrams.
6. What precautions and considerations are important to recognize and manage blood transfusion reactions in patients receiving blood products/artificial blood from JAPAN?
7. Elucidate the lab concept of MGHC, and related blood diseases.
8. Describe the physiology of blood flow regulation to brain, when severe decrease in blood flow happens to patient's brain at Ndola Hitech Hospital.

Section – D – Applied - Answer any FOUR Questions in the white sheets; Marks = 4*6=24

1. A medical student, Mr. Guluguluba, inserted the syringe needle directly in the fourth intercostal space between the ribs and injected a dose of adrenaline into the heart of a dying person at Hillbilly hospital in Twappya.

- i. Describe the mechanism/action of adrenaline (o cardiac muscle).
- ii. How is its physiological mechanism different from Digitalis at the intracellular level?

TERM TWO

August 2025

MBS 210

SECTION

- Starting
- Starting
- Fl

① During each heart beat, only atria contract, then later the ventricle contracts. Explain how is this possible and example diseases if this mechanism fails.

③ A researcher is investigating the physiological basis of different behavioral states in human subjects. They administer various pharmacological agents and observe the resulting changes in physiological responses. List any **FIVE** vasodilators and **FIVE** vasoconstrictors with their effects in various parts of human body.

④ A patient with a history of chest pain presents at Kutwy Hospital. Describe the phases of the cardiac cycle during a typical ventricular arrhythmia, and explain how pressure changes occur in the different heart chambers throughout these phases.

⑤ Human heart has four chambers unlike an octopus, that has two hearts in body. Explain the physiological and clinical significance of coronary angioplasty in managing coronary artery disease, focusing on how blood platelets and atherosclerotic plaques contribute to the blockage of coronary blood flow and how angioplasty addresses these issues.

THE END

SECTION C

23/11/75

- ① - Starling - Frank-Starling law states that the strength of contracting muscle fibres is directly proportional to force like EDV, ESV, the more the heart fills with blood during diastole, the greater the contraction force

Starling forces

i) End ventricular diastole, the volume of blood that remains in the heart at the end of diastole

$$EDV = 120 \text{ mmHg}$$

ii) End ventricular systole, the volume of blood that remains in the ventricle at the end of its contraction

$$ESV = 50 \text{ mmHg}$$

iii) Cardiac output

$$CO =$$

iii) Stroke volume = $EDV - ESV = 70 \text{ mmHg}$

Diseases

Hypertension $EDV > 120 \text{ mmHg}$

myocardial infarction

~~Stroke~~ ^{Arterial depolarization}



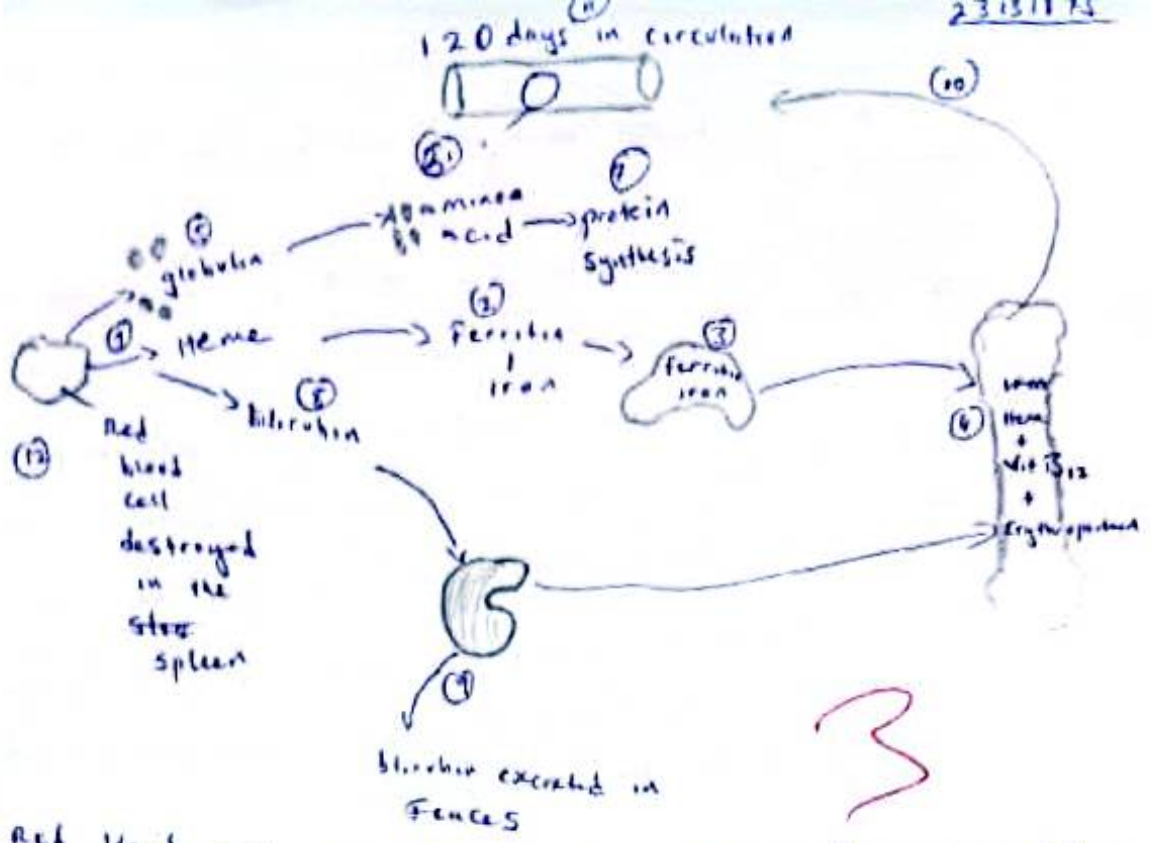
P wave - Arterial depolarization

QRS - complex - ventricular depolarization

T - wave - ventricular repolarization

- i) ST Elevation $\Rightarrow$ elevation of ST segment may in lead II, aVF may mean there is myocardial infarction
- ii) Abnormal Q-wave $\Rightarrow$ may mean there is st. a problem with ventricles
- iii) Diminished P-wave $\Rightarrow$ this means SA node has problems

5



- Red blood cells are destroyed after their 120 day in circulation in the spleen
- Heme and globulin are reused
- They are produced in the red bone marrow
- This is done by negative feedback mechanism

⑦ - Mean corpuscular Hemoglobin concentration this is the concentration of hemoglobin in one erythrocyte

- ranges from 31-37 g/dl
- Related diseases
- Hypochromatic anemia where the concentration is of hemoglobin is low, iron deficiency anemia
- Normochromatic anemia is where the concentration is normal and mean corpuscular volume is normal eg in hemolytic anemia
- Hypochromatic anemia where concentration of hemoglobin is high eg in chronic diseases

2 1/2

b) Vasoconstrictors

Thromboxane $A_2 \Rightarrow$ it works mainly in blood clotting and causes vasoconstriction which leads to platelet aggregation

Epinephrine & norepinephrine $\Rightarrow$ works on beta 1 adrenergic receptor.

5

Link Between Blood Platelet and Atherosclerotic plaques

- Blood platelets may aggregate and form a thrombus which may occlude the coronary artery and prevent blood flow to the heart

• This may be due to i) inflammations of blood vessels

ii) damage to endothelium

iii) chronic disease

- Atherosclerotic plaques, this is the accumulation of plaques in the blood vessels specifically coronary blood vessel and may cause reduction of blood causing cardiac shock.

• It may also dislodge and lodge itself in vital organs such as brain, lung and heart

CORONARY ANGIOPLASTY

- coronary angioplasty is a remedy for the above disorders



- This method requires inserting a balloon in the occluded blood vessels which dilates them and allows excess or physiological blood flow

SECTION D

23/11/25

Control
X
This is possible because stimulation of electrical action potential starts from the sinoatrial node located in the atrium, hence atria contracts first.

- The action potential is stimulated due to increased permeability of ions
- After this it travels to reaches the Atrial ventricular node,
- There is slight delay in reaching AV node due to diminished gap junction
- The Action potential travels through AV to left and right branches untill it reaches the purkinje fibres and the ventricles contract

Pathology

- If this mechanism fails, it must be due to mis firing of the SA node in a condition called arrhythmias

③ i) VASODILATORS

- prostaglandin
- Etor
- Nitric oxide
- Cytochrome
- Viagra eg Sildenafil
- Leukotrienes

ii) Vasoconstrictors

- Thromboxane A₂
- Epinephrine
- Antiplatelet eg Aspirin
- Angiotensin II
- Aldosterone

a) vasodilators

→ prostaglandin; it works on blood vessels and it is responsible for vasodilation during fever

→ Viagra eg Sildenafil; it relaxes the smooth muscles and prevents dysfunction

increase in EVD

and causes us

receptor.

(1) CARDIAC CYCLE

(a) physiological

i) Atrial systole

During normal atrial systole the pressure of the atrium increases, leading to a greater pressure than ventricles hence Atrial ventricular valve opens allow blood to flow while AV valve remains closed

ii) Ventricular Diastole systole

a. Isovolumetric systole

During this stage the changes but the volume of blood stays the same leading to slight increase in ventricular pressure greater than atria, hence AV valve closes & aortic valve is closed as

b. Ventricular Ejection

During this phase volume of blood increases due to passive inflow from atrium and the pressure increases greater than that of pulmonary ~~ventricular aortic valve pulmonary~~ vessels hence aortic valve open

iii) ventricular diastole

a. isovolumetric ventricular diastole

- volume same but pressure changes and the aortic valve closes second heart sound

- note, first was when AV valve closed



(2) PATHOLOGICAL

- But this is not the cases in ventricular arrhythmia, due to missing firing of the SA node. the electrical potential will be slow in the SA node as compared to the AV node, hence

- Hence, this causes ventricles to contract fast, causing

- This premature contraction cause ~~a decrease in cardiac~~

~~not the ECG will increase~~ cause an increase in

- This premature contraction

③ - Anticoagulant like when a patient is undergoing surgery and this it may lead to detrimental effects

- Antiplatelet and other related drugs like aspirin blood flow to some organ, if the patient is injured surgery it may lead to adverse effects

- Antihypertensive such as Angiotensin converting enzyme inhibitor and e.g. sp. eplerenone may be use as it target (RAAS) Renin - Angiotensin I - Angiotensin II - Aldosterone System and prevents cardiovascular crisis

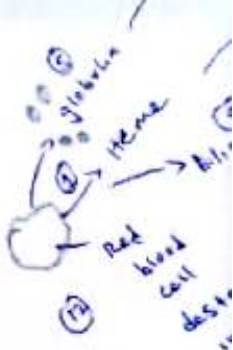
- In conclusion some drugs such as heparin a anticoagulant are contraindicated in patients undergoing surgery

④ - This is called Bara receptor - reflex

- when a person is sitting and stand suddenly stands up there is a slightly drop in pressure due to venous pooling of blood to lower extremities due to gravity

- This is compensated by baro-receptors through vagus and glossopharyngeal nerve found in the aortic arch and carotid sinus

- After wards there will be increase in pressure due to increase in vascular resistance and cardiac output



- increase FVD will increase stroke volume
- This will affect the pressure changes in different chambers of the heart
 - Atrium $\Rightarrow$ low pressure
 - Ventricle $\Rightarrow$ high pressure

- Remedy

- Calcium channel blockers
- Diuretics



⑧

The brain uses myogenic mechanism to control its ^{own} regulation

- when there is decrease in blood through flow to the brain, the cerebral vessels in the brain dilate
- The brain is very sensitive to decrease in blood flow and it may cause the vasomotor system to stimulate nerves to release ~~epinephrine~~ norepinephrine
- α_1 norepinephrine causes the vessels to contract hence increase in vascular resistance resulting in increased blood pressure and blood to the brain



**MEN IN
FLAMES**



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